

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
=====
CONFIGURACIÓN INICIAL
=====
Configuración Valor
      Device  cuda
      Seed    42
Output Width  100
=====
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',           # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 1500,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para cl
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments,
    'extras': {
        'expansion': 32,
        'sparsity_coef': 3e-3,
        'epochs': 1000,
        'patience': 20,
        'learning_rate': 1e-3,
        'sparsity_decay': 0.98,
```

```

        'batch_size': 256
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')

```

```

Metadata de naming:
variante: standard
tecnica: l1-decay
capa: 6
juegos: 1500
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 32, 'sparsity_coef': 0.003, 'epochs': 1000, 'patience': 20, 'learning_rate': 0.0001}

```

```

extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')

```

Nuevo extras registrado: ex2 → {'expansion': 32, 'sparsity_coef': 0.003, 'epochs': 1000, 'patience': 20, 'learning_rate': 0.0001}

extras_id: ex2

3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```

# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 16384,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 256,
    'num_epochs': 1000,
    'learning_rate': 1e-3,
    'sparsity_coef': 3e-3,
    'sparsity_decay': 0.98,

    # Early Stopping
    'early_stopping': True,
    'patience': 20,

    # Datos
    'activations_file': Path(f"../../activations/data/layer{NAMING_META['capa']}_{NAMING_META['juegos']}.npy"),
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}

# Crear directorio local del modelo (plots_dir ya fue creado por get_plots_dir)
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración con formato profesional
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

```

```

# Arquitectura
for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Entrenamiento
for key in ['batch_size', 'num_epochs', 'learning_rate', 'sparsity_coef', 'sparsity_decay']:
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Early Stopping
for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)

```

```

=====
CONFIGURACIÓN DEL SAE
=====

```

Categoría	Parámetro	Valor
Arquitectura	input_dim	512
Arquitectura	hidden_dim	16384
Arquitectura	tied_weights	False
Entrenamiento	batch_size	256
Entrenamiento	num_epochs	1000
Entrenamiento	learning_rate	0.001

```

Entrenamiento sparsity_coef 0.003
Entrenamiento sparsity_decay 0.98
Early Stopping early_stopping True
Early Stopping patience 20
=====

Archivo de activaciones: ..\..\..\activations\data\layer6_1500games.npy
Expansión: 32.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1500games\ex2\plots
=====

```

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})

print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')

```

```

print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

```

```

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False, # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)

```

```

=====
DATASET DE ACTIVACIONES
=====

```

	Propiedad	Valor
	Total muestras	88,500
	Shape (88500, 512)	
	Dtype	float32
Tamaño en memoria		172.85 MB

```

=====

=====
SPLIT DE DATOS (80-20)
=====

```

	Conjunto	Muestras	Porcentaje
	Train	70,800	80.0%
Validation		17,700	20.0%
	Total	88,500	100.0%

```

=====

```

=====				
DATALOADERS				
=====				
Loader	Batches	Batch Size	Shuffle	
Train	277	256	Sí	
Validation	70	256	No	
=====				

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder:** Linear $\rightarrow$ ReLU (sparsity natural)- **Decoder:** Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)

# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim'] # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '_'
    ],
    'Parámetros': [

```

```

        '-',
        f"{encoder_params:,}",
        '-',
        f"{decoder_params:,}",
        '-',
        f"{num_params:,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)

```

```

=====
ARQUITECTURA DEL MODELO
=====

```

	Componente	Dimensión	Parámetros
	Input	512	-
	Encoder	512 → 16384	8,404,992
Hidden	(Latent)	16384	-
	Decoder	16384 → 512	8,389,120
	Output	512	-
	Total	-	16,794,112

```

=====

Factor de expansión: 32.0x
Tied weights: No
Device: cuda
=====

```

6. Función de Pérdida

Loss = MSE(reconstrucción) + λ × L1(activaciones)

- **MSE**: Mide qué tan bien reconstruimos las activaciones originales
- **L1**: Penaliza activaciones grandes → fuerza sparsity

```
def loss_function(x, reconstruction, hidden, sparsity_coef):
    """
    Pérdida compuesta: MSE + L1 penalty

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse)
        sparsity_coef: Peso de la penalización L1
    """
    # Error de reconstrucción
    mse_loss = F.mse_loss(reconstruction, x)

    # Sumar sobre las features (dim=1), promediar sobre el batch
    sparsity_loss = hidden.abs().sum(dim=1).mean()

    # Pérdida total
    total_loss = mse_loss + sparsity_coef * sparsity_loss

    return total_loss, mse_loss, sparsity_loss

# Optimizador
optimizer = torch.optim.Adam(model.parameters(), lr=CONFIG['learning_rate'])

# Resumen del optimizador
optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'Sparsity Decay'],
    'Valor': [
        'Adam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
        f"{CONFIG['sparsity_decay']:.2f}"
    ]
})

print('\n' + '='*50)
print(' OPTIMIZACIÓN')
print('='*50)
print(optim_info.to_string(index=False))
print('='*50)
```

```
=====
OPTIMIZACIÓN
=====
                Parámetro Valor
            Optimizador  Adam
            Learning Rate 1e-03
Sparsity Coef (inicial) 3e-03
            Sparsity Decay 0.98
=====
```

7. Loop de Entrenamiento

```
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
    'sparsity_coef_history': [] # Guardar evolución del coeficiente L1
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' L1 decay: {CONFIG["sparsity_coef"]:.2e} × {CONFIG["sparsity_decay"]:.2f}^época')
print(f' Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):
```

```

# ===== L1 DECAY =====
# Reducir sparsity_coef exponencialmente
current_sparsity_coef = CONFIG['sparsity_coef'] * (CONFIG['sparsity_decay'] ** epoch)
history['sparsity_coef_history'].append(current_sparsity_coef)

# ===== ENTRENAMIENTO =====
model.train()

epoch_total_loss = 0
epoch_mse_loss = 0
epoch_sparsity_loss = 0
epoch_l0 = 0
epoch_fraction_active = 0

# Barra de progreso
pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] L1={current_sparsity_coef}')

for batch in pbar:
    # Mover al device
    x = batch.to(device)

    # Forward pass
    reconstruction, hidden = model(x)

    # Calcular pérdida con sparsity_coef actualizado
    total_loss, mse_loss, sparsity_loss = loss_function(
        x, reconstruction, hidden, current_sparsity_coef
    )

    # Backward pass
    optimizer.zero_grad()
    total_loss.backward()

    optimizer.step()

    # Métricas
    with torch.no_grad():
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

    # Acumular
    epoch_total_loss += total_loss.item()

```

```

epoch_mse_loss += mse_loss.item()
epoch_sparsity_loss += sparsity_loss.item()
epoch_l0 += l0
epoch_fraction_active += fraction_active

# Actualizar barra
pbar.set_postfix({
    'loss': f'{total_loss.item():.4f}',
    'mse': f'{mse_loss.item():.4f}',
    'l0': f'{l0:.1f}'
})

# Promedios de entrenamiento
num_train_batches = len(train_loader)
avg_train_total_loss = epoch_total_loss / num_train_batches
avg_train_mse_loss = epoch_mse_loss / num_train_batches
avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
avg_train_l0 = epoch_l0 / num_train_batches
avg_train_fraction_active = epoch_fraction_active / num_train_batches

# ===== VALIDACIÓN =====
model.eval()

val_total_loss = 0
val_mse_loss = 0
val_sparsity_loss = 0
val_l0 = 0
val_fraction_active = 0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)

        # Forward pass
        reconstruction, hidden = model(x)

        # Calcular pérdida con sparsity_coef actualizado
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, current_sparsity_coef
        )

# Métricas

```

```

10 = (hidden > 0).float().sum(dim=1).mean().item()
fraction_active = (hidden > 0).float().mean().item()

# Acumular
val_total_loss += total_loss.item()
val_mse_loss += mse_loss.item()
val_sparsity_loss += sparsity_loss.item()
val_l0 += l0
val_fraction_active += fraction_active

# Promedios de validación
num_val_batches = len(val_loader)
avg_val_total_loss = val_total_loss / num_val_batches
avg_val_mse_loss = val_mse_loss / num_val_batches
avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
avg_val_l0 = val_l0 / num_val_batches
avg_val_fraction_active = val_fraction_active / num_val_batches

# Guardar en historial
history['train_total_loss'].append(avg_train_total_loss)
history['train_mse_loss'].append(avg_train_mse_loss)
history['train_sparsity_loss'].append(avg_train_sparsity_loss)
history['train_l0_sparsity'].append(avg_train_l0)
history['train_fraction_active'].append(avg_train_fraction_active)

history['val_total_loss'].append(avg_val_total_loss)
history['val_mse_loss'].append(avg_val_mse_loss)
history['val_sparsity_loss'].append(avg_val_sparsity_loss)
history['val_l0_sparsity'].append(avg_val_l0)
history['val_fraction_active'].append(avg_val_fraction_active)

# ===== EARLY STOPPING =====
if CONFIG['early_stopping']:
    if avg_val_mse_loss < best_val_mse:
        best_val_mse = avg_val_mse_loss
        best_epoch = epoch + 1
        epochs_no_improve = 0

    # Guardar mejor modelo (local + organizado)
    best_checkpoint = {
        'model_state_dict': model.state_dict(),
        'optimizer_state_dict': optimizer.state_dict(),

```

```

        'config': CONFIG,
        'history': history,
        'epoch': best_epoch,
        'val_mse': best_val_mse
    }

    # Guardado local
    best_model_name = get_model_name(
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best'
    )
    torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

    # Guardado organizado
    save_checkpoint(
        model,
        base_path=NAMING_META['base_path'],
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=CONFIG,
        history=history,
        optimizer=optimizer
    )

    print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= CONFIG['patience']:
        print(f'\n Early stopping activado en época {epoch+1}')
        print(f' Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}')
        break

# Log cada 5 épocas
if (epoch + 1) % 5 == 0 or epoch == 0:
    print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | L1 coef: {current_sparsity_coef:}')

```

```

        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}')
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0')
        if CONFIG['early_stopping']:
            print(f'    Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {best_val_mse:.4f}')

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

# Tabla de métricas finales
final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})

print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)

```

Iniciando entrenamiento: 1000 épocas máximo
 L1 decay: $3.00e-03 \times 0.98^{\text{época}}$
 Early stopping: patience=20 épocas

=====

Época 1/1000 [Train] L1=3.00e-03: 100%| | 277/277 [00:03<00:00, 72.86it/s, loss=0.5084,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 1 (Val MSE: 0.4311)

Época 1/1000 | L1 coef: 3.00e-03

Train - Total: 0.6425 | MSE: 0.5556 | L0: 124.5

Val - Total: 0.4964 | MSE: 0.4311 | L0: 37.0

Best Val MSE: 0.4311 @ época 1 | Sin mejora: 0/20

Época 2/1000 [Train] L1=2.94e-03: 100%| | 277/277 [00:03<00:00, 78.85it/s, loss=0.3899,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 2 (Val MSE: 0.3072)

Época 3/1000 [Train] L1=2.88e-03: 100%| | 277/277 [00:03<00:00, 79.13it/s, loss=0.3526,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 3 (Val MSE: 0.2571)

Época 4/1000 [Train] L1=2.82e-03: 100%| | 277/277 [00:03<00:00, 77.91it/s, loss=0.3138,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 4 (Val MSE: 0.2266)

Época 5/1000 [Train] L1=2.77e-03: 100%| | 277/277 [00:03<00:00, 78.74it/s, loss=0.2774,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 5 (Val MSE: 0.2053)

Época 5/1000 | L1 coef: 2.77e-03

Train - Total: 0.3032 | MSE: 0.2123 | L0: 130.0

Val - Total: 0.2953 | MSE: 0.2053 | L0: 139.1

Best Val MSE: 0.2053 @ época 5 | Sin mejora: 0/20

Época 6/1000 [Train] L1=2.71e-03: 100%| | 277/277 [00:03<00:00, 79.12it/s, loss=0.2777,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 6 (Val MSE: 0.1909)

Época 7/1000 [Train] L1=2.66e-03: 100%| | 277/277 [00:03<00:00, 78.27it/s, loss=0.2677,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 7 (Val MSE: 0.1791)

Época 8/1000 [Train] L1=2.60e-03: 100%| | 277/277 [00:03<00:00, 78.62it/s, loss=0.2509,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 8 (Val MSE: 0.1675)

Época 9/1000 [Train] L1=2.55e-03: 100%| | 277/277 [00:03<00:00, 79.48it/s, loss=0.2391,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 9 (Val MSE: 0.1565)

Época 10/1000 [Train] L1=2.50e-03: 100%| | 277/277 [00:02<00:00, 92.69it/s, loss=0.2266,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 10 (Val MSE: 0.1489)

Época 10/1000 | L1 coef: 2.50e-03

Train - Total: 0.2309 | MSE: 0.1503 | L0: 191.9

Val - Total: 0.2291 | MSE: 0.1489 | L0: 198.5

Best Val MSE: 0.1489 @ época 10 | Sin mejora: 0/20

Época 11/1000 [Train] L1=2.45e-03: 100%| | 277/277 [00:03<00:00, 90.64it/s, loss=0.2155,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 11 (Val MSE: 0.1407)

Época 12/1000 [Train] L1=2.40e-03: 100%| | 277/277 [00:03<00:00, 79.09it/s, loss=0.2121,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 12 (Val MSE: 0.1337)

Época 13/1000 [Train] L1=2.35e-03: 100%| | 277/277 [00:03<00:00, 77.31it/s, loss=0.2008,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 13 (Val MSE: 0.1305)

Época 14/1000 [Train] L1=2.31e-03: 100%| | 277/277 [00:03<00:00, 81.73it/s, loss=0.1912,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 14 (Val MSE: 0.1254)

Época 15/1000 [Train] L1=2.26e-03: 100%| | 277/277 [00:03<00:00, 76.99it/s, loss=0.1900,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 15 (Val MSE: 0.1191)

Época 15/1000 | L1 coef: 2.26e-03

Train - Total: 0.1901 | MSE: 0.1186 | L0: 234.7

Val - Total: 0.1903 | MSE: 0.1191 | L0: 239.8

Best Val MSE: 0.1191 @ época 15 | Sin mejora: 0/20

Época 16/1000 [Train] L1=2.22e-03: 100%| | 277/277 [00:02<00:00, 92.61it/s, loss=0.1849,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 16 (Val MSE: 0.1168)

Época 17/1000 [Train] L1=2.17e-03: 100%| | 277/277 [00:03<00:00, 90.48it/s, loss=0.1759,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 17 (Val MSE: 0.1125)

Época 18/1000 [Train] L1=2.13e-03: 100%| | 277/277 [00:03<00:00, 79.69it/s, loss=0.1721,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 18 (Val MSE: 0.1102)

Época 19/1000 [Train] L1=2.09e-03: 100%| | 277/277 [00:03<00:00, 80.45it/s, loss=0.1722,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 19 (Val MSE: 0.1077)

Época 20/1000 [Train] L1=2.04e-03: 100%| | 277/277 [00:03<00:00, 82.76it/s, loss=0.1648,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 20 (Val MSE: 0.1058)

Época 20/1000 | L1 coef: 2.04e-03

Train - Total: 0.1670 | MSE: 0.1047 | L0: 259.2

Val - Total: 0.1679 | MSE: 0.1058 | L0: 262.5

Best Val MSE: 0.1058 @ época 20 | Sin mejora: 0/20

Época 21/1000 [Train] L1=2.00e-03: 100%| | 277/277 [00:03<00:00, 79.47it/s, loss=0.1627,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 21 (Val MSE: 0.1049)

Época 22/1000 [Train] L1=1.96e-03: 100%| | 277/277 [00:03<00:00, 83.39it/s, loss=0.1613,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 22 (Val MSE: 0.1021)

Época 23/1000 [Train] L1=1.92e-03: 100%| | 277/277 [00:03<00:00, 88.71it/s, loss=0.1647,

Época 24/1000 [Train] L1=1.89e-03: 100%| | 277/277 [00:03<00:00, 89.99it/s, loss=0.1538,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 24 (Val MSE: 0.0995)

Época 25/1000 [Train] L1=1.85e-03: 100%| | 277/277 [00:03<00:00, 87.05it/s, loss=0.1554,

Época 25/1000 | L1 coef: 1.85e-03
Train - Total: 0.1540 | MSE: 0.0995 | L0: 272.7
Val - Total: 0.1575 | MSE: 0.1034 | L0: 273.2
Best Val MSE: 0.0995 @ época 24 | Sin mejora: 1/20

Época 26/1000 [Train] L1=1.81e-03: 100%| | 277/277 [00:03<00:00, 79.22it/s, loss=0.1454,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 26 (Val MSE: 0.0952)

Época 27/1000 [Train] L1=1.77e-03: 100%| | 277/277 [00:03<00:00, 79.59it/s, loss=0.1489,
Época 28/1000 [Train] L1=1.74e-03: 100%| | 277/277 [00:03<00:00, 79.70it/s, loss=0.1504,
Época 29/1000 [Train] L1=1.70e-03: 100%| | 277/277 [00:03<00:00, 79.25it/s, loss=0.1398,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 29 (Val MSE: 0.0937)

Época 30/1000 [Train] L1=1.67e-03: 100%| | 277/277 [00:03<00:00, 78.23it/s, loss=0.1378,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 30 (Val MSE: 0.0925)

Época 30/1000 | L1 coef: 1.67e-03
Train - Total: 0.1399 | MSE: 0.0909 | L0: 285.6
Val - Total: 0.1414 | MSE: 0.0925 | L0: 288.2
Best Val MSE: 0.0925 @ época 30 | Sin mejora: 0/20

Época 31/1000 [Train] L1=1.64e-03: 100%| | 277/277 [00:03<00:00, 77.52it/s, loss=0.1324,
Época 32/1000 [Train] L1=1.60e-03: 100%| | 277/277 [00:03<00:00, 78.53it/s, loss=0.1365,
Época 33/1000 [Train] L1=1.57e-03: 100%| | 277/277 [00:03<00:00, 78.78it/s, loss=0.1351,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 33 (Val MSE: 0.0917)

Época 34/1000 [Train] L1=1.54e-03: 100%| | 277/277 [00:03<00:00, 78.77it/s, loss=0.1350,

Época 35/1000 [Train] L1=1.51e-03: 100%| | 277/277 [00:03<00:00, 83.29it/s, loss=0.1331,

Época 35/1000 | L1 coef: 1.51e-03

Train - Total: 0.1312 | MSE: 0.0868 | L0: 294.9

Val - Total: 0.1368 | MSE: 0.0927 | L0: 295.1

Best Val MSE: 0.0917 @ época 33 | Sin mejora: 2/20

Época 36/1000 [Train] L1=1.48e-03: 100%| | 277/277 [00:03<00:00, 84.75it/s, loss=0.1324,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 36 (Val MSE: 0.0909)

Época 37/1000 [Train] L1=1.45e-03: 100%| | 277/277 [00:03<00:00, 92.12it/s, loss=0.1274,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 37 (Val MSE: 0.0866)

Época 38/1000 [Train] L1=1.42e-03: 100%| | 277/277 [00:03<00:00, 81.43it/s, loss=0.1217,

Época 39/1000 [Train] L1=1.39e-03: 100%| | 277/277 [00:03<00:00, 77.10it/s, loss=0.1213,

Época 40/1000 [Train] L1=1.36e-03: 100%| | 277/277 [00:03<00:00, 79.47it/s, loss=0.1295,

Época 40/1000 | L1 coef: 1.36e-03

Train - Total: 0.1246 | MSE: 0.0845 | L0: 300.3

Val - Total: 0.1319 | MSE: 0.0921 | L0: 300.6

Best Val MSE: 0.0866 @ época 37 | Sin mejora: 3/20

Época 41/1000 [Train] L1=1.34e-03: 100%| | 277/277 [00:03<00:00, 80.28it/s, loss=0.1217,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 41 (Val MSE: 0.0847)

Época 42/1000 [Train] L1=1.31e-03: 100%| | 277/277 [00:03<00:00, 81.17it/s, loss=0.1231,
Época 43/1000 [Train] L1=1.28e-03: 100%| | 277/277 [00:03<00:00, 81.73it/s, loss=0.1226,
Época 44/1000 [Train] L1=1.26e-03: 100%| | 277/277 [00:03<00:00, 79.29it/s, loss=0.1181,
Época 45/1000 [Train] L1=1.23e-03: 100%| | 277/277 [00:03<00:00, 78.53it/s, loss=0.1176,

Época 45/1000 | L1 coef: 1.23e-03

Train - Total: 0.1190 | MSE: 0.0826 | L0: 304.8

Val - Total: 0.1220 | MSE: 0.0857 | L0: 305.9

Best Val MSE: 0.0847 @ época 41 | Sin mejora: 4/20

Época 46/1000 [Train] L1=1.21e-03: 100%| | 277/277 [00:03<00:00, 79.24it/s, loss=0.1202,
Época 47/1000 [Train] L1=1.18e-03: 100%| | 277/277 [00:03<00:00, 80.69it/s, loss=0.1179,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 47 (Val MSE: 0.0840)

Época 48/1000 [Train] L1=1.16e-03: 100%| | 277/277 [00:03<00:00, 78.44it/s, loss=0.1104,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 48 (Val MSE: 0.0807)

Época 49/1000 [Train] L1=1.14e-03: 100%| | 277/277 [00:03<00:00, 79.43it/s, loss=0.1168,
Época 50/1000 [Train] L1=1.11e-03: 100%| | 277/277 [00:03<00:00, 79.44it/s, loss=0.1154,

Época 50/1000 | L1 coef: 1.11e-03

Train - Total: 0.1152 | MSE: 0.0820 | L0: 308.5

Val - Total: 0.1192 | MSE: 0.0862 | L0: 308.5

Best Val MSE: 0.0807 @ época 48 | Sin mejora: 2/20

Época 51/1000 [Train] L1=1.09e-03: 100%| | 277/277 [00:03<00:00, 79.29it/s, loss=0.1196,
Época 52/1000 [Train] L1=1.07e-03: 100%| | 277/277 [00:03<00:00, 79.95it/s, loss=0.1132,
Época 53/1000 [Train] L1=1.05e-03: 100%| | 277/277 [00:03<00:00, 81.20it/s, loss=0.1109,
Época 54/1000 [Train] L1=1.03e-03: 100%| | 277/277 [00:03<00:00, 80.64it/s, loss=0.1110,
Época 55/1000 [Train] L1=1.01e-03: 100%| | 277/277 [00:03<00:00, 79.95it/s, loss=0.1140,

Época 55/1000 | L1 coef: 1.01e-03
Train - Total: 0.1115 | MSE: 0.0809 | L0: 312.0
Val - Total: 0.1135 | MSE: 0.0828 | L0: 313.1
Best Val MSE: 0.0807 @ época 48 | Sin mejora: 7/20

Época 56/1000 [Train] L1=9.88e-04: 100%| | 277/277 [00:03<00:00, 88.49it/s, loss=0.1143,
Época 57/1000 [Train] L1=9.68e-04: 100%| | 277/277 [00:03<00:00, 78.01it/s, loss=0.1049,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 57 (Val MSE: 0.0798)

Época 58/1000 [Train] L1=9.48e-04: 100%| | 277/277 [00:03<00:00, 78.67it/s, loss=0.1053,
Época 59/1000 [Train] L1=9.29e-04: 100%| | 277/277 [00:03<00:00, 77.71it/s, loss=0.1052,
Época 60/1000 [Train] L1=9.11e-04: 100%| | 277/277 [00:03<00:00, 79.08it/s, loss=0.1019,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 60 (Val MSE: 0.0792)

Época 60/1000 | L1 coef: 9.11e-04
Train - Total: 0.1072 | MSE: 0.0790 | L0: 315.1
Val - Total: 0.1075 | MSE: 0.0792 | L0: 317.3
Best Val MSE: 0.0792 @ época 60 | Sin mejora: 0/20

Época 61/1000 [Train] L1=8.93e-04: 100%| | 277/277 [00:03<00:00, 79.02it/s, loss=0.1094,
Época 62/1000 [Train] L1=8.75e-04: 100%| | 277/277 [00:03<00:00, 80.55it/s, loss=0.1063,
Época 63/1000 [Train] L1=8.57e-04: 100%| | 277/277 [00:03<00:00, 79.02it/s, loss=0.1038,
Época 64/1000 [Train] L1=8.40e-04: 100%| | 277/277 [00:03<00:00, 78.55it/s, loss=0.1057,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 64 (Val MSE: 0.0791)

Época 65/1000 [Train] L1=8.23e-04: 100%| | 277/277 [00:03<00:00, 78.67it/s, loss=0.1026,

Época 65/1000 | L1 coef: 8.23e-04
Train - Total: 0.1024 | MSE: 0.0761 | L0: 319.4
Val - Total: 0.1059 | MSE: 0.0796 | L0: 320.5
Best Val MSE: 0.0791 @ época 64 | Sin mejora: 1/20

Época 66/1000 [Train] L1=8.07e-04: 100%| | 277/277 [00:03<00:00, 80.19it/s, loss=0.1055,
Época 67/1000 [Train] L1=7.91e-04: 100%| | 277/277 [00:03<00:00, 78.61it/s, loss=0.1035,
Época 68/1000 [Train] L1=7.75e-04: 100%| | 277/277 [00:03<00:00, 78.77it/s, loss=0.1020,
Época 69/1000 [Train] L1=7.59e-04: 100%| | 277/277 [00:03<00:00, 80.21it/s, loss=0.0996,
Época 70/1000 [Train] L1=7.44e-04: 100%| | 277/277 [00:03<00:00, 79.37it/s, loss=0.0959

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 70 (Val MSE: 0.0760)

Época 70/1000 | L1 coef: 7.44e-04
Train - Total: 0.0997 | MSE: 0.0755 | L0: 321.8
Val - Total: 0.1003 | MSE: 0.0760 | L0: 323.0
Best Val MSE: 0.0760 @ época 70 | Sin mejora: 0/20

Época 71/1000 [Train] L1=7.29e-04: 100%| | 277/277 [00:03<00:00, 78.27it/s, loss=0.1007,
Época 72/1000 [Train] L1=7.15e-04: 100%| | 277/277 [00:03<00:00, 78.82it/s, loss=0.1010,
Época 73/1000 [Train] L1=7.00e-04: 100%| | 277/277 [00:03<00:00, 77.97it/s, loss=0.0983,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 73 (Val MSE: 0.0750)

Época 74/1000 [Train] L1=6.86e-04: 100%| | 277/277 [00:03<00:00, 80.74it/s, loss=0.0986,
Época 75/1000 [Train] L1=6.73e-04: 100%| | 277/277 [00:03<00:00, 79.15it/s, loss=0.0991,

Época 75/1000 | L1 coef: 6.73e-04
Train - Total: 0.0982 | MSE: 0.0759 | L0: 323.2
Val - Total: 0.1013 | MSE: 0.0790 | L0: 323.7
Best Val MSE: 0.0750 @ época 73 | Sin mejora: 2/20

Época 76/1000 [Train] L1=6.59e-04: 100%| | 277/277 [00:03<00:00, 79.12it/s, loss=0.0951,
Época 77/1000 [Train] L1=6.46e-04: 100%| | 277/277 [00:03<00:00, 79.55it/s, loss=0.0984,
Época 78/1000 [Train] L1=6.33e-04: 100%| | 277/277 [00:03<00:00, 78.40it/s, loss=0.0971,
Época 79/1000 [Train] L1=6.21e-04: 100%| | 277/277 [00:03<00:00, 79.11it/s, loss=0.0911,
Época 80/1000 [Train] L1=6.08e-04: 100%| | 277/277 [00:03<00:00, 78.44it/s, loss=0.0946,

Época 80/1000 | L1 coef: 6.08e-04
Train - Total: 0.0936 | MSE: 0.0728 | L0: 326.1
Val - Total: 0.0967 | MSE: 0.0758 | L0: 327.6
Best Val MSE: 0.0750 @ época 73 | Sin mejora: 7/20

Época 81/1000 [Train] L1=5.96e-04: 100%| | 277/277 [00:03<00:00, 79.62it/s, loss=0.0935,
Época 82/1000 [Train] L1=5.84e-04: 100%| | 277/277 [00:03<00:00, 79.45it/s, loss=0.0939,
Época 83/1000 [Train] L1=5.72e-04: 100%| | 277/277 [00:03<00:00, 78.92it/s, loss=0.0905,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 83 (Val MSE: 0.0746)

Época 84/1000 [Train] L1=5.61e-04: 100%| | 277/277 [00:03<00:00, 79.24it/s, loss=0.0935,
Época 85/1000 [Train] L1=5.50e-04: 100%| | 277/277 [00:03<00:00, 78.96it/s, loss=0.1010,

Época 85/1000 | L1 coef: 5.50e-04
Train - Total: 0.0946 | MSE: 0.0755 | L0: 326.2
Val - Total: 0.0963 | MSE: 0.0770 | L0: 328.7
Best Val MSE: 0.0746 @ época 83 | Sin mejora: 2/20

Época 86/1000 [Train] L1=5.39e-04: 100%| | 277/277 [00:03<00:00, 77.84it/s, loss=0.0940,
Época 87/1000 [Train] L1=5.28e-04: 100%| | 277/277 [00:03<00:00, 78.89it/s, loss=0.0918,
Época 88/1000 [Train] L1=5.17e-04: 100%| | 277/277 [00:03<00:00, 78.84it/s, loss=0.0946,
Época 89/1000 [Train] L1=5.07e-04: 100%| | 277/277 [00:03<00:00, 81.44it/s, loss=0.0915,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 89 (Val MSE: 0.0738)

Época 90/1000 [Train] L1=4.97e-04: 100%| | 277/277 [00:03<00:00, 90.65it/s, loss=0.0932,

Época 90/1000 | L1 coef: 4.97e-04
Train - Total: 0.0891 | MSE: 0.0711 | L0: 329.3
Val - Total: 0.0925 | MSE: 0.0746 | L0: 329.3
Best Val MSE: 0.0738 @ época 89 | Sin mejora: 1/20

Época 91/1000 [Train] L1=4.87e-04: 100%| | 277/277 [00:03<00:00, 80.23it/s, loss=0.0901,
Época 92/1000 [Train] L1=4.77e-04: 100%| | 277/277 [00:03<00:00, 78.99it/s, loss=0.0891,
Época 93/1000 [Train] L1=4.68e-04: 100%| | 277/277 [00:03<00:00, 80.25it/s, loss=0.0888,
Época 94/1000 [Train] L1=4.58e-04: 100%| | 277/277 [00:03<00:00, 80.08it/s, loss=0.0870,
Época 95/1000 [Train] L1=4.49e-04: 100%| | 277/277 [00:03<00:00, 79.61it/s, loss=0.0879

Época 95/1000 | L1 coef: 4.49e-04

Train - Total: 0.0920 | MSE: 0.0754 | L0: 328.7

Val - Total: 0.0918 | MSE: 0.0752 | L0: 331.9

Best Val MSE: 0.0738 @ época 89 | Sin mejora: 6/20

Época 96/1000 [Train] L1=4.40e-04: 100%| | 277/277 [00:03<00:00, 79.86it/s, loss=0.0872,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 96 (Val MSE: 0.0718)

Época 97/1000 [Train] L1=4.31e-04: 100%| | 277/277 [00:03<00:00, 90.51it/s, loss=0.0859,
Época 98/1000 [Train] L1=4.23e-04: 100%| | 277/277 [00:03<00:00, 81.01it/s, loss=0.0871,
Época 99/1000 [Train] L1=4.14e-04: 100%| | 277/277 [00:03<00:00, 78.79it/s, loss=0.0895,
Época 100/1000 [Train] L1=4.06e-04: 100%| | 277/277 [00:03<00:00, 79.25it/s, loss=0.0871

Época 100/1000 | L1 coef: 4.06e-04

Train - Total: 0.0886 | MSE: 0.0732 | L0: 330.7

Val - Total: 0.0900 | MSE: 0.0745 | L0: 332.7

Best Val MSE: 0.0718 @ época 96 | Sin mejora: 4/20

Época 101/1000 [Train] L1=3.98e-04: 100%| | 277/277 [00:03<00:00, 81.96it/s, loss=0.0867
Época 102/1000 [Train] L1=3.90e-04: 100%| | 277/277 [00:03<00:00, 90.98it/s, loss=0.0838

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 102 (Val MSE: 0.0716)

Época 103/1000 [Train] L1=3.82e-04: 100%| | 277/277 [00:03<00:00, 83.09it/s, loss=0.0902
Época 104/1000 [Train] L1=3.74e-04: 100%| | 277/277 [00:03<00:00, 86.66it/s, loss=0.0818

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 104 (Val MSE: 0.0704)

Época 105/1000 [Train] L1=3.67e-04: 100%| | 277/277 [00:03<00:00, 91.10it/s, loss=0.0855]

Época 105/1000 | L1 coef: 3.67e-04
Train - Total: 0.0838 | MSE: 0.0694 | L0: 333.2
Val - Total: 0.0863 | MSE: 0.0719 | L0: 334.8
Best Val MSE: 0.0704 @ época 104 | Sin mejora: 1/20

Época 106/1000 [Train] L1=3.60e-04: 100%| | 277/277 [00:02<00:00, 93.15it/s, loss=0.0881]
Época 107/1000 [Train] L1=3.52e-04: 100%| | 277/277 [00:02<00:00, 92.70it/s, loss=0.0811]
Época 108/1000 [Train] L1=3.45e-04: 100%| | 277/277 [00:03<00:00, 92.29it/s, loss=0.0834]
Época 109/1000 [Train] L1=3.38e-04: 100%| | 277/277 [00:02<00:00, 92.64it/s, loss=0.0856]
Época 110/1000 [Train] L1=3.32e-04: 100%| | 277/277 [00:02<00:00, 92.50it/s, loss=0.0805]

Época 110/1000 | L1 coef: 3.32e-04
Train - Total: 0.0837 | MSE: 0.0704 | L0: 333.7
Val - Total: 0.0868 | MSE: 0.0736 | L0: 333.4
Best Val MSE: 0.0704 @ época 104 | Sin mejora: 6/20

Época 111/1000 [Train] L1=3.25e-04: 100%| | 277/277 [00:02<00:00, 92.91it/s, loss=0.0830]
Época 112/1000 [Train] L1=3.19e-04: 100%| | 277/277 [00:02<00:00, 92.42it/s, loss=0.0823]
Época 113/1000 [Train] L1=3.12e-04: 100%| | 277/277 [00:02<00:00, 92.34it/s, loss=0.0849]
Época 114/1000 [Train] L1=3.06e-04: 100%| | 277/277 [00:02<00:00, 93.34it/s, loss=0.0782]

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 114 (Val MSE: 0.0693)

Época 115/1000 [Train] L1=3.00e-04: 100%| | 277/277 [00:03<00:00, 91.26it/s, loss=0.0828]

Época 115/1000 | L1 coef: 3.00e-04
Train - Total: 0.0810 | MSE: 0.0686 | L0: 335.2
Val - Total: 0.0833 | MSE: 0.0708 | L0: 336.3
Best Val MSE: 0.0693 @ época 114 | Sin mejora: 1/20

Época 116/1000 [Train] L1=2.94e-04: 100%| | 277/277 [00:02<00:00, 92.66it/s, loss=0.0808
Época 117/1000 [Train] L1=2.88e-04: 100%| | 277/277 [00:02<00:00, 92.48it/s, loss=0.0806
Época 118/1000 [Train] L1=2.82e-04: 100%| | 277/277 [00:02<00:00, 92.83it/s, loss=0.0972
Época 119/1000 [Train] L1=2.77e-04: 100%| | 277/277 [00:02<00:00, 93.18it/s, loss=0.0790

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 119 (Val MSE: 0.0693)

Época 120/1000 [Train] L1=2.71e-04: 100%| | 277/277 [00:03<00:00, 92.13it/s, loss=0.0803

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 120 (Val MSE: 0.0692)

Época 120/1000 | L1 coef: 2.71e-04
Train - Total: 0.0801 | MSE: 0.0685 | L0: 336.0
Val - Total: 0.0808 | MSE: 0.0692 | L0: 337.2
Best Val MSE: 0.0692 @ época 120 | Sin mejora: 0/20

Época 121/1000 [Train] L1=2.66e-04: 100%| | 277/277 [00:03<00:00, 91.89it/s, loss=0.0801
Época 122/1000 [Train] L1=2.60e-04: 100%| | 277/277 [00:03<00:00, 90.56it/s, loss=0.0797
Época 123/1000 [Train] L1=2.55e-04: 100%| | 277/277 [00:03<00:00, 92.30it/s, loss=0.0840
Época 124/1000 [Train] L1=2.50e-04: 100%| | 277/277 [00:02<00:00, 92.60it/s, loss=0.0776
Época 125/1000 [Train] L1=2.45e-04: 100%| | 277/277 [00:03<00:00, 92.33it/s, loss=0.0788

Época 125/1000 | L1 coef: 2.45e-04
Train - Total: 0.0812 | MSE: 0.0706 | L0: 335.9
Val - Total: 0.0817 | MSE: 0.0709 | L0: 338.2
Best Val MSE: 0.0692 @ época 120 | Sin mejora: 5/20

Época 126/1000 [Train] L1=2.40e-04: 100%| | 277/277 [00:02<00:00, 92.59it/s, loss=0.0799
Época 127/1000 [Train] L1=2.35e-04: 100%| | 277/277 [00:02<00:00, 92.73it/s, loss=0.0767
Época 128/1000 [Train] L1=2.31e-04: 100%| | 277/277 [00:03<00:00, 90.01it/s, loss=0.0809
Época 129/1000 [Train] L1=2.26e-04: 100%| | 277/277 [00:03<00:00, 92.14it/s, loss=0.0792

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 129 (Val MSE: 0.0683)

Época 130/1000 [Train] L1=2.21e-04: 100%| | 277/277 [00:03<00:00, 91.20it/s, loss=0.0769

Época 130/1000 | L1 coef: 2.21e-04

Train - Total: 0.0796 | MSE: 0.0697 | L0: 336.9

Val - Total: 0.0803 | MSE: 0.0703 | L0: 338.6

Best Val MSE: 0.0683 @ época 129 | Sin mejora: 1/20

Época 131/1000 [Train] L1=2.17e-04: 100%| | 277/277 [00:03<00:00, 91.06it/s, loss=0.0746

Época 132/1000 [Train] L1=2.13e-04: 100%| | 277/277 [00:03<00:00, 91.15it/s, loss=0.0775

Época 133/1000 [Train] L1=2.08e-04: 100%| | 277/277 [00:03<00:00, 91.96it/s, loss=0.0765

Época 134/1000 [Train] L1=2.04e-04: 100%| | 277/277 [00:02<00:00, 92.50it/s, loss=0.0727

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 134 (Val MSE: 0.0678)

Época 135/1000 [Train] L1=2.00e-04: 100%| | 277/277 [00:03<00:00, 91.18it/s, loss=0.0790

Época 135/1000 | L1 coef: 2.00e-04

Train - Total: 0.0773 | MSE: 0.0681 | L0: 338.1

Val - Total: 0.0780 | MSE: 0.0687 | L0: 340.0

Best Val MSE: 0.0678 @ época 134 | Sin mejora: 1/20

Época 136/1000 [Train] L1=1.96e-04: 100%| | 277/277 [00:02<00:00, 92.41it/s, loss=0.0808

Época 137/1000 [Train] L1=1.92e-04: 100%| | 277/277 [00:03<00:00, 91.53it/s, loss=0.0782

Época 138/1000 [Train] L1=1.88e-04: 100%| | 277/277 [00:03<00:00, 91.30it/s, loss=0.0735

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 138 (Val MSE: 0.0670)

Época 139/1000 [Train] L1=1.85e-04: 100%| | 277/277 [00:03<00:00, 90.10it/s, loss=0.0742

Época 140/1000 [Train] L1=1.81e-04: 100%| | 277/277 [00:02<00:00, 92.49it/s, loss=0.0788

Época 140/1000 | L1 coef: 1.81e-04

Train - Total: 0.0770 | MSE: 0.0684 | L0: 338.5

Val - Total: 0.0807 | MSE: 0.0722 | L0: 338.2

Best Val MSE: 0.0670 @ época 138 | Sin mejora: 2/20

Época 141/1000 [Train] L1=1.77e-04: 100%| | 277/277 [00:03<00:00, 91.98it/s, loss=0.0771
 Época 142/1000 [Train] L1=1.74e-04: 100%| | 277/277 [00:03<00:00, 91.27it/s, loss=0.0746
 Época 143/1000 [Train] L1=1.70e-04: 100%| | 277/277 [00:03<00:00, 91.69it/s, loss=0.0759
 Época 144/1000 [Train] L1=1.67e-04: 100%| | 277/277 [00:03<00:00, 92.25it/s, loss=0.0803
 Época 145/1000 [Train] L1=1.64e-04: 100%| | 277/277 [00:03<00:00, 91.74it/s, loss=0.0750

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
 Mejor modelo guardado en época 145 (Val MSE: 0.0661)

Época 145/1000 | L1 coef: 1.64e-04
 Train - Total: 0.0748 | MSE: 0.0668 | L0: 339.6
 Val - Total: 0.0742 | MSE: 0.0661 | L0: 341.8
 Best Val MSE: 0.0661 @ época 145 | Sin mejora: 0/20

Época 146/1000 [Train] L1=1.60e-04: 100%| | 277/277 [00:03<00:00, 91.74it/s, loss=0.0707
 Época 147/1000 [Train] L1=1.57e-04: 100%| | 277/277 [00:03<00:00, 92.04it/s, loss=0.0745
 Época 148/1000 [Train] L1=1.54e-04: 100%| | 277/277 [00:03<00:00, 92.31it/s, loss=0.0764
 Época 149/1000 [Train] L1=1.51e-04: 100%| | 277/277 [00:02<00:00, 93.19it/s, loss=0.0766
 Época 150/1000 [Train] L1=1.48e-04: 100%| | 277/277 [00:03<00:00, 91.75it/s, loss=0.0718

Época 150/1000 | L1 coef: 1.48e-04
 Train - Total: 0.0731 | MSE: 0.0657 | L0: 340.4
 Val - Total: 0.0744 | MSE: 0.0669 | L0: 341.1
 Best Val MSE: 0.0661 @ época 145 | Sin mejora: 5/20

Época 151/1000 [Train] L1=1.45e-04: 100%| | 277/277 [00:03<00:00, 91.81it/s, loss=0.0719
 Época 152/1000 [Train] L1=1.42e-04: 100%| | 277/277 [00:03<00:00, 92.15it/s, loss=0.0720
 Época 153/1000 [Train] L1=1.39e-04: 100%| | 277/277 [00:02<00:00, 92.43it/s, loss=0.0778
 Época 154/1000 [Train] L1=1.36e-04: 100%| | 277/277 [00:03<00:00, 92.15it/s, loss=0.0714
 Época 155/1000 [Train] L1=1.34e-04: 100%| | 277/277 [00:03<00:00, 91.38it/s, loss=0.0729

Época 155/1000 | L1 coef: 1.34e-04
 Train - Total: 0.0705 | MSE: 0.0636 | L0: 341.6
 Val - Total: 0.0749 | MSE: 0.0680 | L0: 342.4
 Best Val MSE: 0.0661 @ época 145 | Sin mejora: 10/20

Época 156/1000 [Train] L1=1.31e-04: 100%| | 277/277 [00:03<00:00, 91.91it/s, loss=0.0688
Época 157/1000 [Train] L1=1.28e-04: 100%| | 277/277 [00:03<00:00, 91.58it/s, loss=0.0733
Época 158/1000 [Train] L1=1.26e-04: 100%| | 277/277 [00:02<00:00, 92.68it/s, loss=0.0750
Época 159/1000 [Train] L1=1.23e-04: 100%| | 277/277 [00:03<00:00, 91.56it/s, loss=0.0711
Época 160/1000 [Train] L1=1.21e-04: 100%| | 277/277 [00:03<00:00, 92.05it/s, loss=0.0719

Época 160/1000 | L1 coef: 1.21e-04

Train - Total: 0.0736 | MSE: 0.0672 | L0: 340.9

Val - Total: 0.0743 | MSE: 0.0679 | L0: 341.9

Best Val MSE: 0.0661 @ época 145 | Sin mejora: 15/20

Época 161/1000 [Train] L1=1.18e-04: 100%| | 277/277 [00:03<00:00, 92.22it/s, loss=0.0748
Época 162/1000 [Train] L1=1.16e-04: 100%| | 277/277 [00:02<00:00, 93.07it/s, loss=0.0711
Época 163/1000 [Train] L1=1.14e-04: 100%| | 277/277 [00:03<00:00, 91.97it/s, loss=0.0706

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 163 (Val MSE: 0.0655)

Época 164/1000 [Train] L1=1.11e-04: 100%| | 277/277 [00:03<00:00, 91.52it/s, loss=0.0704
Época 165/1000 [Train] L1=1.09e-04: 100%| | 277/277 [00:02<00:00, 92.71it/s, loss=0.0733

Época 165/1000 | L1 coef: 1.09e-04

Train - Total: 0.0722 | MSE: 0.0662 | L0: 341.5

Val - Total: 0.0739 | MSE: 0.0680 | L0: 342.2

Best Val MSE: 0.0655 @ época 163 | Sin mejora: 2/20

Época 166/1000 [Train] L1=1.07e-04: 100%| | 277/277 [00:03<00:00, 91.93it/s, loss=0.0690
Época 167/1000 [Train] L1=1.05e-04: 100%| | 277/277 [00:02<00:00, 93.14it/s, loss=0.0701
Época 168/1000 [Train] L1=1.03e-04: 100%| | 277/277 [00:02<00:00, 93.10it/s, loss=0.0665
Época 169/1000 [Train] L1=1.01e-04: 100%| | 277/277 [00:03<00:00, 92.00it/s, loss=0.0754
Época 170/1000 [Train] L1=9.87e-05: 100%| | 277/277 [00:03<00:00, 91.96it/s, loss=0.0715

Época 170/1000 | L1 coef: 9.87e-05

Train - Total: 0.0709 | MSE: 0.0653 | L0: 342.2

Val - Total: 0.0713 | MSE: 0.0657 | L0: 343.1

Best Val MSE: 0.0655 @ época 163 | Sin mejora: 7/20

Época 171/1000 [Train] L1=9.67e-05: 100%| | 277/277 [00:02<00:00, 93.46it/s, loss=0.0724
 Época 172/1000 [Train] L1=9.48e-05: 100%| | 277/277 [00:03<00:00, 92.09it/s, loss=0.0701
 Época 173/1000 [Train] L1=9.29e-05: 100%| | 277/277 [00:03<00:00, 92.17it/s, loss=0.0683
 Época 174/1000 [Train] L1=9.10e-05: 100%| | 277/277 [00:03<00:00, 91.98it/s, loss=0.0693
 Época 175/1000 [Train] L1=8.92e-05: 100%| | 277/277 [00:03<00:00, 90.75it/s, loss=0.0726

Época 175/1000 | L1 coef: 8.92e-05
 Train - Total: 0.0698 | MSE: 0.0646 | L0: 342.7
 Val - Total: 0.0721 | MSE: 0.0670 | L0: 343.9
 Best Val MSE: 0.0655 @ época 163 | Sin mejora: 12/20

Época 176/1000 [Train] L1=8.74e-05: 100%| | 277/277 [00:03<00:00, 92.25it/s, loss=0.0703
 Época 177/1000 [Train] L1=8.57e-05: 100%| | 277/277 [00:03<00:00, 91.78it/s, loss=0.0693

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
 Mejor modelo guardado en época 177 (Val MSE: 0.0652)

Época 178/1000 [Train] L1=8.40e-05: 100%| | 277/277 [00:03<00:00, 91.46it/s, loss=0.0709
 Época 179/1000 [Train] L1=8.23e-05: 100%| | 277/277 [00:03<00:00, 91.67it/s, loss=0.0729
 Época 180/1000 [Train] L1=8.06e-05: 100%| | 277/277 [00:03<00:00, 92.02it/s, loss=0.0685

Época 180/1000 | L1 coef: 8.06e-05
 Train - Total: 0.0693 | MSE: 0.0645 | L0: 343.0
 Val - Total: 0.0705 | MSE: 0.0657 | L0: 344.6
 Best Val MSE: 0.0652 @ época 177 | Sin mejora: 3/20

Época 181/1000 [Train] L1=7.90e-05: 100%| | 277/277 [00:02<00:00, 93.04it/s, loss=0.0691
 Época 182/1000 [Train] L1=7.75e-05: 100%| | 277/277 [00:03<00:00, 92.08it/s, loss=0.0697
 Época 183/1000 [Train] L1=7.59e-05: 100%| | 277/277 [00:02<00:00, 92.55it/s, loss=0.0695

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
 Mejor modelo guardado en época 183 (Val MSE: 0.0650)

Época 184/1000 [Train] L1=7.44e-05: 100%| | 277/277 [00:03<00:00, 92.00it/s, loss=0.0704
 Época 185/1000 [Train] L1=7.29e-05: 100%| | 277/277 [00:02<00:00, 92.76it/s, loss=0.0675

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 185 (Val MSE: 0.0641)

Época 185/1000 | L1 coef: 7.29e-05
Train - Total: 0.0689 | MSE: 0.0645 | L0: 343.3
Val - Total: 0.0685 | MSE: 0.0641 | L0: 344.9
Best Val MSE: 0.0641 @ época 185 | Sin mejora: 0/20

Época 186/1000 [Train] L1=7.14e-05: 100%| | 277/277 [00:02<00:00, 92.70it/s, loss=0.0732
Época 187/1000 [Train] L1=7.00e-05: 100%| | 277/277 [00:03<00:00, 89.88it/s, loss=0.0653

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt
Mejor modelo guardado en época 187 (Val MSE: 0.0634)

Época 188/1000 [Train] L1=6.86e-05: 100%| | 277/277 [00:03<00:00, 90.50it/s, loss=0.0650
Época 189/1000 [Train] L1=6.72e-05: 100%| | 277/277 [00:03<00:00, 91.51it/s, loss=0.0706
Época 190/1000 [Train] L1=6.59e-05: 100%| | 277/277 [00:03<00:00, 91.48it/s, loss=0.0690

Época 190/1000 | L1 coef: 6.59e-05
Train - Total: 0.0686 | MSE: 0.0645 | L0: 343.6
Val - Total: 0.0689 | MSE: 0.0648 | L0: 345.0
Best Val MSE: 0.0634 @ época 187 | Sin mejora: 3/20

Época 191/1000 [Train] L1=6.46e-05: 100%| | 277/277 [00:03<00:00, 92.30it/s, loss=0.0684
Época 192/1000 [Train] L1=6.33e-05: 100%| | 277/277 [00:02<00:00, 93.00it/s, loss=0.0700
Época 193/1000 [Train] L1=6.20e-05: 100%| | 277/277 [00:03<00:00, 91.90it/s, loss=0.0675
Época 194/1000 [Train] L1=6.08e-05: 100%| | 277/277 [00:03<00:00, 91.77it/s, loss=0.0637
Época 195/1000 [Train] L1=5.96e-05: 100%| | 277/277 [00:03<00:00, 91.22it/s, loss=0.0653

Época 195/1000 | L1 coef: 5.96e-05
Train - Total: 0.0671 | MSE: 0.0633 | L0: 344.2
Val - Total: 0.0674 | MSE: 0.0636 | L0: 345.7
Best Val MSE: 0.0634 @ época 187 | Sin mejora: 8/20

Época 196/1000 [Train] L1=5.84e-05: 100%| | 277/277 [00:02<00:00, 93.18it/s, loss=0.0702
Época 197/1000 [Train] L1=5.72e-05: 100%| | 277/277 [00:03<00:00, 92.10it/s, loss=0.0688
Época 198/1000 [Train] L1=5.61e-05: 100%| | 277/277 [00:03<00:00, 91.74it/s, loss=0.0672
Época 199/1000 [Train] L1=5.49e-05: 100%| | 277/277 [00:03<00:00, 91.55it/s, loss=0.0698
Época 200/1000 [Train] L1=5.38e-05: 100%| | 277/277 [00:03<00:00, 91.08it/s, loss=0.0671

Época 200/1000 | L1 coef: 5.38e-05

Train - Total: 0.0663 | MSE: 0.0628 | L0: 344.6

Val - Total: 0.0672 | MSE: 0.0637 | L0: 345.5

Best Val MSE: 0.0634 @ época 187 | Sin mejora: 13/20

Época 201/1000 [Train] L1=5.28e-05: 100%| | 277/277 [00:03<00:00, 92.33it/s, loss=0.0687
Época 202/1000 [Train] L1=5.17e-05: 100%| | 277/277 [00:03<00:00, 90.93it/s, loss=0.0670
Época 203/1000 [Train] L1=5.07e-05: 100%| | 277/277 [00:03<00:00, 91.51it/s, loss=0.0659
Época 204/1000 [Train] L1=4.97e-05: 100%| | 277/277 [00:03<00:00, 92.25it/s, loss=0.0684
Época 205/1000 [Train] L1=4.87e-05: 100%| | 277/277 [00:03<00:00, 91.83it/s, loss=0.0649

Época 205/1000 | L1 coef: 4.87e-05

Train - Total: 0.0691 | MSE: 0.0659 | L0: 344.0

Val - Total: 0.0681 | MSE: 0.0648 | L0: 345.3

Best Val MSE: 0.0634 @ época 187 | Sin mejora: 18/20

Época 206/1000 [Train] L1=4.77e-05: 100%| | 277/277 [00:02<00:00, 92.95it/s, loss=0.0644

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_best.pt

Mejor modelo guardado en época 206 (Val MSE: 0.0622)

Época 207/1000 [Train] L1=4.67e-05: 100%| | 277/277 [00:03<00:00, 90.69it/s, loss=0.0652
Época 208/1000 [Train] L1=4.58e-05: 100%| | 277/277 [00:03<00:00, 89.30it/s, loss=0.0650
Época 209/1000 [Train] L1=4.49e-05: 100%| | 277/277 [00:03<00:00, 91.42it/s, loss=0.0637
Época 210/1000 [Train] L1=4.40e-05: 100%| | 277/277 [00:03<00:00, 91.30it/s, loss=0.0625

Época 210/1000 | L1 coef: 4.40e-05

Train - Total: 0.0666 | MSE: 0.0636 | L0: 344.8

Val - Total: 0.0675 | MSE: 0.0645 | L0: 346.0

Best Val MSE: 0.0622 @ época 206 | Sin mejora: 4/20

Época 211/1000 [Train] L1=4.31e-05: 100%| | 277/277 [00:03<00:00, 91.32it/s, loss=0.0655
 Época 212/1000 [Train] L1=4.22e-05: 100%| | 277/277 [00:03<00:00, 91.70it/s, loss=0.0649
 Época 213/1000 [Train] L1=4.14e-05: 100%| | 277/277 [00:03<00:00, 91.47it/s, loss=0.0651
 Época 214/1000 [Train] L1=4.06e-05: 100%| | 277/277 [00:03<00:00, 90.87it/s, loss=0.0654
 Época 215/1000 [Train] L1=3.98e-05: 100%| | 277/277 [00:03<00:00, 91.03it/s, loss=0.0661

Época 215/1000 | L1 coef: 3.98e-05
 Train - Total: 0.0650 | MSE: 0.0623 | L0: 345.3
 Val - Total: 0.0671 | MSE: 0.0643 | L0: 346.5
 Best Val MSE: 0.0622 @ época 206 | Sin mejora: 9/20

Época 216/1000 [Train] L1=3.90e-05: 100%| | 277/277 [00:02<00:00, 92.76it/s, loss=0.0651
 Época 217/1000 [Train] L1=3.82e-05: 100%| | 277/277 [00:03<00:00, 91.65it/s, loss=0.0648
 Época 218/1000 [Train] L1=3.74e-05: 100%| | 277/277 [00:03<00:00, 91.80it/s, loss=0.0668
 Época 219/1000 [Train] L1=3.67e-05: 100%| | 277/277 [00:03<00:00, 91.10it/s, loss=0.0635
 Época 220/1000 [Train] L1=3.59e-05: 100%| | 277/277 [00:03<00:00, 90.97it/s, loss=0.0662

Época 220/1000 | L1 coef: 3.59e-05
 Train - Total: 0.0641 | MSE: 0.0615 | L0: 345.7
 Val - Total: 0.0676 | MSE: 0.0650 | L0: 346.6
 Best Val MSE: 0.0622 @ época 206 | Sin mejora: 14/20

Época 221/1000 [Train] L1=3.52e-05: 100%| | 277/277 [00:03<00:00, 90.98it/s, loss=0.0643
 Época 222/1000 [Train] L1=3.45e-05: 100%| | 277/277 [00:03<00:00, 92.21it/s, loss=0.0630
 Época 223/1000 [Train] L1=3.38e-05: 100%| | 277/277 [00:03<00:00, 90.39it/s, loss=0.0650
 Época 224/1000 [Train] L1=3.32e-05: 100%| | 277/277 [00:03<00:00, 91.07it/s, loss=0.0645
 Época 225/1000 [Train] L1=3.25e-05: 100%| | 277/277 [00:03<00:00, 91.10it/s, loss=0.0699

Época 225/1000 | L1 coef: 3.25e-05
 Train - Total: 0.0652 | MSE: 0.0629 | L0: 345.4
 Val - Total: 0.0687 | MSE: 0.0663 | L0: 345.4
 Best Val MSE: 0.0622 @ época 206 | Sin mejora: 19/20

Época 226/1000 [Train] L1=3.18e-05: 100%| | 277/277 [00:03<00:00, 91.08it/s, loss=0.0611

Early stopping activado en época 226
Mejor modelo: época 206 con Val MSE = 0.0622

```
=====
RESUMEN DEL ENTRENAMIENTO
=====
Mejor modelo en época 206 con Val MSE = 0.0622
Early stopping tras 226 épocas

Métricas Finales (Última Época):
      Métrica  Train Validation
Total Loss 0.0647      0.0651
MSE Loss 0.0624      0.0628
LO Sparsity 345.6      347.0
Fraction Active 2.11%    2.12%
=====
```

8. Guardar Modelo

```
# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final'
)
checkpoint_path = CONFIG['save_dir'] / final_model_name
torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
```

```

model,
base_path=NAMING_META['base_path'],
variante=NAMING_META['variante'],
tecnica=NAMING_META['tecnica'],
capa=NAMING_META['capa'],
juegos=NAMING_META['juegos'],
estado='final',
extras=NAMING_META['extras'],
config=CONFIG,
history=history,
optimizer=optimizer
)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)

```

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1500games\sae_standard_l1-decay_l6_1500g_ex2_final.pt

```

=====
CHECKPOINTS GUARDADOS
=====

```

Tipo	Ubicación
Local	saved_model\sae_standard_l1-decay_l6_1500g_final.pt
Organizado	layer_06/models/...

```

=====

```

9. Métricas básicas del modelo

```
# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "sparsity_decay": CONFIG["sparsity_decay"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

# Resumen
metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",

```

```

        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})
print(" " + "="*50)
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)

```

```

=====
METRICAS DE ENTRENAMIENTO GUARDADAS
=====

```

	Metrica	Valor
Val MSE (final)	0.062795	
Val MSE (mejor)	0.062184	
Val L0 (final)	347.0	
Fraccion Activa	2.12%	
Archivo:	c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1500games\ex2\metrics\training_metrics.json	

```

=====

```

10. Visualización de Resultados

```

# Gráficas de pérdidas (Train vs Validation) + L1 Decay
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)

```

```

if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best Epoch')
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8)
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0.8)
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# Sparsity Coefficient (L1 Decay)
axes[1, 1].plot(history['sparsity_coef_history'], color='brown', linewidth=2)
axes[1, 1].set_title('L1 Coefficient (Decay)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('Sparsity Coef')
axes[1, 1].set_yscale('log')
axes[1, 1].grid(True, alpha=0.3)

# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best Epoch')
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')

```

```

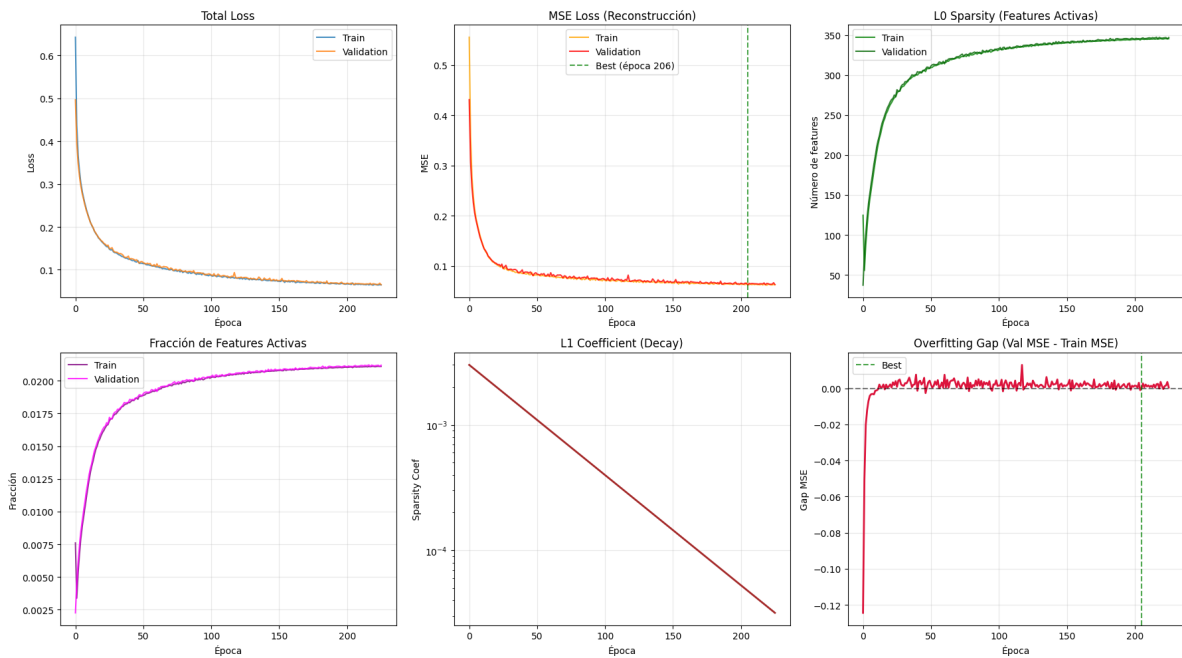
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('  Gráficas guardadas exitosamente')

print('='*50)

```



=====

Gráficas guardadas exitosamente

=====

11. Resumen Final

```

# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:}"
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):}",
        f"{len(val_dataset):}",
        f"{len(full_dataset):}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')

```

```

print(final_summary.to_string(index=False))

# Archivos
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)
print('  ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)

```

```

=====
RESUMEN FINAL DEL ENTRENAMIENTO
=====

```

Arquitectura:

Componente	Valor
Input	512
Hidden	16384 (32.0x)
Parámetros	16,794,112

Datos:

Conjunto	Muestras
Train	70,800
Validation	17,700
Total	88,500

Métricas Finales:

Métrica	Valor
Train Loss	0.0647
Val Loss	0.0651
Train MSE	0.0624
Val MSE	0.0628
Train L0	345.6
Val L0	347.0

Archivos Guardados:

Tipo

Modelo

decay_l6_1500g_final.pt

Gráficas c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard

decay_standard\1500games\ex2\plots\training_curves.png

```
=====
ENTRENAMIENTO COMPLETADO EXITOSAMENTE
=====
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_l1-decay.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
```

```

print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1500games\ex2\sae_standard_l1-decay_l6_1500g_ex2_training.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1500games\ex2\sae_standard_l1-decay_l6_1500g_ex2_training.pdf